A Sharper Degree Bound for the Algebraic Reduction

Abstract. This companion note sharpens the bounded-remainder argument used in *Rank-Two Module SVP in Power-of-Two Cyclotomic Rings*. A Vandermonde bound improves the reciprocal-resultant estimate, and a degree budget controls divided derivatives even when the auxiliary polynomial has repeated factors. For coefficient height at most L^2, the required extension degree decreases from $O(L^3)$ to $O(L^2 \log L)$. We give all proofs. The main paper retains the shorter argument based only on triangle inequalities; the present refinement is optional.

1 A reciprocal-resultant bound

For a monic integer polynomial p of degree d with roots $\alpha_1, \dots, \alpha_d$, write

$$M(p) = \prod_i \max(1, |\alpha_i|).$$

When $p(0) = \pm 1$, its normalized reciprocal is the monic integer polynomial $p^{\#}(X) = X^d p(1/X)/p(0)$.

Lemma 1. *If $p(0) = \pm 1$ and $p \neq p^{\#}$, then $\log M(p) \geq 1/(12d)$.*

Proof. Put $T = \log M(p)$. Both power sums $s_m = \sum_i \alpha_i^m$ and $s_{-m} = \sum_i \alpha_i^{-m}$ are integers. Newton's identities give $s_m \neq s_{-m}$ for some $1 \leq m \leq d$. Write $\alpha_i = e^{u_i + \mathrm{i}\theta_i}$. Since $|p(0)| = 1$, we have $\sum_i |u_i| = 2T$ and $|u_i| \leq T$. Conjugation permutes the roots, hence

$$1 \leq |s_m - s_{-m}| \leq \sum_i |e^{mu_i} - e^{-mu_i}| \leq 2me^{mT} \sum_i |u_i| = 4me^{mT}T.$$

If $T \geq 1/d$, the claim follows. Otherwise $e^{mT} < 3$, giving $T \geq 1/(12d)$.

Lemma 2 (Reciprocal resultant). *Suppose $p(0) = \pm 1$ and $\gcd(p, p^{\#}) = 1$. For every integer $D \geq 1$,*

$$\left|\mathrm{Res}(p, 1 + X^D)\right| \geq (2d)^{-d} M(p)^{D-2d}.$$

Proof. Put $r_i = |\alpha_i|$. Order the roots of $p^{\#}$ as $1/\overline{\alpha}_i$. Since the resultant is a nonzero integer,

$$1 \leq \left|\mathrm{Res}(p, p^{\#})\right| = \prod_{i,j} \left|\alpha_i - \frac{1}{\overline{\alpha}_j}\right|. \tag{1}$$

We bound its off-diagonal factors by introducing

$$V(z_1, \dots, z_d) = \prod_{i<j} |1 - z_i \overline{z}_j|.$$

For each fixed choice of the other coordinates, V is a constant times the modulus of the polynomial $\prod_{j \neq i}(1 - z_i \overline{z}_j)$ in z_i, of degree at most $d-1$. This is a statement in each coordinate separately; no joint holomorphicity is asserted.

For any polynomial f of degree at most k, the maximum-modulus principle gives

$$|f(z)| \leq \max(1, |z|)^k \max_{|u|=1} |f(u)|.$$

For $|z| > 1$, apply the same principle to $u^k f(1/u)$. Applying this inequality successively in all coordinates yields

$$V(z_1, \dots, z_d) \leq \prod_i \max(1, |z_i|)^{d-1} \max_{|u_1|=\cdots=|u_d|=1} V(u_1, \dots, u_d).$$

On the unit circle, V is the absolute value of the Vandermonde determinant $\det(u_i^{j-1})_{i,j}$. Each row has Euclidean norm $\sqrt{d}$, so Hadamard's inequality gives

$$V(\alpha_1, \dots, \alpha_d) \leq d^{d/2} M(p)^{d-1}. \tag{2}$$

Since $\prod_i r_i = 1$, the off-diagonal product in (1) is exactly $V(\alpha_1, \dots, \alpha_d)^2$. Consequently

$$\prod_i |r_i - r_i^{-1}| \geq d^{-d} M(p)^{-2(d-1)}. \tag{3}$$

For each root, the reverse triangle inequality gives

$$|1 + \alpha_i^D| \geq \max(1, r_i)^D \frac{|r_i - r_i^{-1}|}{2 \max(r_i, r_i^{-1})}.$$

Indeed, for $r_i \geq 1$ use $1 - r_i^{-D} \geq 1 - r_i^{-1}$; for $r_i \leq 1$ use $1 - r_i^D \geq 1 - r_i$. Finally $\prod_i \max(r_i, r_i^{-1}) = M(p)^2$. Multiplying and applying (3) proves the claim.

2 Bounded remainders, including repeated factors

Repeated factors only require divided derivatives. Their order and the field degree share the same degree budget, which avoids an exponential coefficient estimate.

Theorem 1. *Let $w \in \mathbb{Z}[X]$ be monic of degree $L \geq 2$, with $w(0) = \pm 1$, and suppose no irreducible factor of w is reciprocal. Let $H \geq 1$ and put $A = \mathbb{Z}[X]/(w)$. If*

$$D \geq 16 L^2 \left\lceil \log_2(2HL^2) \right\rceil,$$

then every $e \in \mathbb{Z}[X]$ satisfying

$$\deg e < L, \qquad \|e\|_\infty \leq H, \qquad e \in (1 + X^D)A$$

is zero. The polynomial w may have repeated factors.

Proof. Suppose $e \neq 0$, and lift the membership relation to

$$e = (1 + X^D)q + wu, \qquad q, u \in \mathbb{Z}[X].$$

Some irreducible factor p of w satisfies $j := \operatorname{ord}_p(e) < r := \operatorname{ord}_p(w)$, since $\deg e < L$. Write $d = \deg p$ and let α be a root of p. The constant term of p is ± 1, and p is nonreciprocal. Thus $\gcd(p, p^{\#}) = 1$ and p has no unit-circle root: such a root, together with its conjugate, would give a common root of $p, p^{\#}$. In particular $p \nmid 1 + X^D$, so $\operatorname{ord}_p(q) = j$.

Set $e_j = e^{(j)}/j!$ and $q_j = q^{(j)}/j!$. These are integer polynomials. At α, the derivatives of q below order j and those of wu through order j vanish. The jth divided derivative therefore gives

$$e_j(\alpha) = (1 + \alpha^D)q_j(\alpha) \neq 0. \tag{4}$$

The nonzero algebraic integer $q_j(\alpha)$ has norm of absolute value at least one.

Each coefficient of e_j has absolute value at most $H\binom{L-1}{j} \leq HL^j$. Evaluation at the d roots of p gives

$$|N(e_j(\alpha))| \leq (HL^{j+1})^d M(p)^{L-1}.$$

Combining (4) with lemma 2 yields

$$(2d)^{-d} M(p)^{D-2d} \leq (HL^{j+1})^d M(p)^{L-1}.$$

Since $(j+1)d \leq rd \leq L$, taking logarithms and applying lemma 1 gives

$$\begin{aligned} \frac{D - 2d - L + 1}{12d} &\leq d \log(2d) + d \log H + d(j+1) \log L \\ &\leq L \log(2HL^2). \end{aligned}$$

Put $h = \lceil \log_2(2HL^2) \rceil$. The right side is less than Lh, whereas $D \geq 16L^2h$ makes the left side at least $4Lh/3 - 1/4 > Lh$, a contradiction.

Corollary 1. *For $H = L^2$ and $L \geq 2$, the conclusion of theorem 1 holds whenever*

$$D \geq 64L^2 \lceil \log_2 L \rceil.$$

The least dyadic degree satisfying this condition is less than $128L^2 \lceil \log_2 L \rceil$.

Proof. Put $\ell = \lceil \log_2 L \rceil \geq 1$. The last comparison in the theorem's proof has right side $L \log(2L^4) < L(1 + 4\ell)$. The asserted threshold makes its left side at least $16L\ell/3 - 1/4 > L(1 + 4\ell)$. Rounding up to a power of two costs a factor strictly below two.

The corollary changes only the source-dependent extension degree in the main reduction. The coefficient packing and the universal degree threshold for the rigid anchor remain as in the main paper.